

SHUNYA

Corrected Evolution Implementation Plan

2026-07-14 — Revision with 6 Master Directive Corrections

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SHUNYA CORRECTED EVOLUTION IMPLEMENTATION PLAN

Phase Order (per Master Directive + Corrections)

0 → 1 → 2 → 3 → 3a → 4 → 5 → 6 → 7 → 8 → 9 → 10 → 11 → 12 → 13 → 14 → 15 → 16 → 17

Correction: Phase 3a (Memory Architecture) inserted between Phase 3 (Conversation) and Phase 4 (Human Context).

PHASE 0 — Safety, Baseline & Architectural Convergence

Same as previous plan. No correction required.

PHASE 1 — Unified Human Identity Domain

Corrected Model:

Person - id (UUID) - tenant_id - canonical_name — full legal/official name - preferred_name — how they want to be addressed - primary_language - locale - timezone - identity_status (ACTIVE, MERGED, DECEASED, ARCHIVED) - created_at - updated_at

PersonIdentity - id - person_id (FK) - identity_type: email | phone | whatsapp | telegram | application_account | external_channel - identity_value - normalized_value - verification_state: UNVERIFIED | EXACT | VERIFIED | PROBABLE | AMBIGUOUS | UNRESOLVED

Role Profiles (role/business projections over Person — NOT separate identity): - EmployeeProfile: tenant_id, person_id, employee_code, department, manager_person_id, role, status, joined_at - CustomerProfile: tenant_id, person_id, preferred_channel, communication_style, lifetime_value, segment - SupplierContactProfile: tenant_id, person_id, supplier_id, role_in_organization, is_primary - ClientUserProfile: tenant_id, person_id, portal_access_granted, last_login

These are projections, NOT redefinitions of human identity.

PHASE 2 — Relationship Intelligence Domain

Corrected Model:

Relationship - id (UUID) - tenant_id - subject_entity_type: person | organization | team - subject_entity_id (UUID) - relationship_type: EMPLOYEE_OF | OWNS | CUSTOMER_OF | SUPPLIER_OF | COLLEAGUE_OF | FAMILY_OF | REFERRED_BY - object_entity_type: person | organization | team - object_entity_id (UUID) - status: ACTIVE | INACTIVE | SUSPENDED | ENDED - confidence (0.0–1.0) - source_type: EXPLICIT | INFERRED | IMPORTED | SYSTEM_DERIVED - source_provenance — link to Evidence record - metadata (JSONB) - valid_from - valid_until - created_at - updated_at

The runtime MUST structurally distinguish: - EXPLICIT RELATIONSHIP FACT (source_type=EXPLICIT) - INFERRED RELATIONSHIP SIGNAL (source_type=INFERRED)

Never silently promote inference into fact. Confidence < 1.0 + source_type=INFERRED requires evidence accumulation before promotion.

PHASE 3 — Conversation & Interaction Domain

Corrected Model:

Conversation - id (UUID) - tenant_id - conversation_type: inquiry | support | internal | system | booking | followup - status: ACTIVE | PAUSED | RESOLVED | ARCHIVED - primary_person_id (FK → Person) - primary_organization_id (FK → Tenant) - subject — short label - context_entity_type — what business object this is about - context_entity_id - started_at - last_activity_at - metadata (JSONB)

ConversationParticipant - id - conversation_id (FK) - person_id (FK) — NULL for system - participant_type: human | ai | system | bot - channel: whatsapp | telegram | in_app | email | voice | system - joined_at - left_at

Interaction / Message - id (UUID) - conversation_id (FK) - actor_type: person | ai | system - actor_person_id (FK → Person, nullable) - channel: whatsapp | telegram | in_app | email | voice | system - direction: inbound | outbound - message_type: text | image | document | template | action | system_event - content (Text) - content_rendered (HTML/rich — nullable) - reply_to_interaction_id (FK → self, nullable) - external_message_id — WhatsApp msg ID, Telegram msg ID, etc. - channel_metadata (JSONB) - sensitivity: PUBLIC | ORGANIZATIONAL | PERSONAL | SENSITIVE | HIGHLY_RESTRICTED - created_at

ConversationContext - id - conversation_id (FK) - entity_type: person | plan | booking | lead | invoice | task - entity_id (UUID) - entity_label — human-readable ("her Bali trip") - resolution_state: ACTIVE | RESOLVED | STALE - activated_at - resolved_at

Mapping existing channels into unified model:

WhatsApp webhook (/whatsapp/webhook): → handle_whatsapp_incoming() → resolve sender phone → PersonIdentity → Person → find or create Conversation (by sender Person + tenant) → create Interaction(channel=whatsapp, direction=inbound, external_message_id=msg.id) → update Conversation.last_activity_at

Telegram webhook (/telegram/webhook): → same pattern, channel=telegram

ShunyaInterface.process_message(): → same pattern, channel=in_app (or whatsapp/telegram depending on caller)

WhatsAppAdapter.send(): → creates Interaction(channel=whatsapp, direction=outbound, external_message_id=response.id)

PHASE 3a — MEMORY ARCHITECTURE (NEW — Correction #1)

Seven memory classes with distinct ownership, persistence, retrieval, retention, and integration:

WORKING MEMORY

- Ownership: Conversation session
- Persistence: In-process (Alpine.js store + server session cache, TTL)
- Retrieval: Current conversation scope only
- Retention: Session lifetime; discarded on conversation close
- Integration: ContextFusionEngine reads working memory before reasoning
- Backing store: Redis (in-memory) when available; database `working_memory` table with TTL

CONVERSATION MEMORY

- Ownership: Conversation domain
- Persistence: Database (interactions table)

- Retrieval: By conversation_id; scoped to active conversations
- Retention: TTL based on sensitivity + conversation status
- Integration: ContextFusionEngine loads last N interactions per conversation
- Backing store: interactions table (already part of Phase 3)

RELATIONSHIP MEMORY

- Ownership: Relationship domain
- Persistence: Database (relationships table)
- Retrieval: By person_id + relationship_type
- Retention: Relationship lifecycle (status=ACTIVE)
- Integration: ContextFusionEngine loads eligible relationships for reasoning context
- Backing store: relationships table (Phase 2)

ORGANIZATIONAL MEMORY

- Ownership: Knowledge domain
- Persistence: KnowledgeStore (knowledge_facts table)
- Retrieval: By domain + category + fact_key
- Retention: Immutable with versioning; sensitivity-gated
- Integration: KnowledgeLayer before reasoning; sufficiency evaluation
- Backing store: knowledge_facts table (EXISTS, PRESERVED)

HUMAN PREFERENCE MEMORY

- Ownership: Human Context domain (Phase 4)
- Persistence: Database (pattern_candidates → promoted to human_signals with signal_type=preference)
- Retrieval: By person_id + signal_type
- Retention: Temporal validity window; sensitivity-gated
- Integration: ContextFusionEngine loads eligible preferences
- Backing store: human_signals table (Phase 4)

EPISODIC MEMORY

- Ownership: Observation/Learning domain
- Persistence: Database (observations, learning_entries tables)
- Retrieval: By entity_id + episode_type; pattern matching
- Retention: TTL; summarization for long-term retention
- Integration: LearningLayer promotes episodic patterns into organizational memory
- Backing store: observations table (EXISTS, PRESERVED)

WORLD INTELLIGENCE MEMORY

- Ownership: World Intelligence domain (Phase 11)
- Persistence: Database (world_events, retrieved_sources tables)
- Retrieval: By relevance (organization, role, topic)
- Retention: Freshness-based eviction; relevance-gated persistence
- Integration: ContextFusionEngine loads world context only when freshness-sensitive
- Backing store: world_intelligence tables (Phase 11)

Universal Memory Metadata (every durable memory item): - provenance (link to Source/Evidence record) - temporal_validity (valid_from, valid_until) - confidence (0.0-1.0) - sensitivity (PUBLIC|ORGANIZATIONAL|PERSONAL|SENSITIVE|HIGHLY_RESTRICTED) - retention_policy (link to RetentionPolicy) - visibility (SELF|TEAM|ORGANIZATION|GLOBAL) - memory_class (WORKING|CONVERSATION|RELATIONSHIP|ORGANIZATIONAL|PREFERENCE|EPISODIC|WORLD) - supersession (link to superseding memory record, if any)

PHASE 4 — Human Context Domain

No structural correction. Phase 4 creates HumanSignal and PatternCandidate. Corrected sensitivity classification applied (see Phase 5).

PHASE 5 — Privacy, Sensitivity & Memory Eligibility

Corrected Sensitivity Classification:

- **PUBLIC** — No restriction; may be exposed to any user, any context
- **ORGANIZATIONAL** — Organization-owned information; visible within tenant
- **PERSONAL** — Human-personal context voluntarily shared; structurally distinct from organization-owned information; requires consent tracking
- **SENSITIVE** — Requires explicit authorization; health, political, financial details
- **HIGHLY_RESTRICTED** — Limited to specific authorized roles; audit-logged access

PERSONAL is restored. SHUNYA must structurally distinguish human-personal context from organization-owned information. A Person's preference for "I prefer window seats" is PERSONAL. The organization's policy "All business-class bookings get window seats when available" is ORGANIZATIONAL. These are stored separately and serve different reasoning contexts.

Memory Eligibility boundaries (reconciled with Phase 3a): - CAN_UNDERSTAND_NOW → Working Memory - CAN_RETAIN_IN_SESSION → Working Memory, Conversation Memory - CAN_RETAIN_TEMPORARILY → Conversation Memory, Episodic Memory - CAN_REMEMBER_LONG_TERM → Organizational Memory, Human Preference Memory - CAN_EXPOSE_TO_ORGANIZATION → Organizational Memory (sensitivity ≤ ORGANIZATIONAL) - CAN_USE_FOR_BUSINESS_REASONING → Organizational Memory, Relationship Memory - CAN_USE_FOR_COMMERCIAL_ACTION → Requires explicit consent record - CAN_SHARE_WITH_ANOTHER_EMPLOYEE → Sensitivity ≤ ORGANIZATIONAL + visibility ≥ TEAM - CAN_DELETE_FORGET → Deletion API with audit trail

PHASE 6 — Evidence, Provenance & Source Intelligence

Inserted after Memory Architecture (Phase 3a) per correction. Same as previous plan.

PHASE 7 — Source/Evidence/Fact/Inference Runtime

Inserted after Phase 6. Same as previous plan.

PHASE 8 — LLM Intelligence Runtime

Inserted after Phase 7. Same as previous plan.

PHASE 9 — Context Fusion Engine

Inserted after Phase 8. Updated integration path:

Corrected flow after all phases:

```
ShunyaInterface.process_message()  
→ ContextFusionEngine  
  → HUMAN_CONTEXT (Phase 4, gated by Phase 5)  
  → ORGANIZATIONAL_CONTEXT (Phase 3a – Organizational Memory)  
  → RELATIONSHIP_CONTEXT (Phase 2 + Phase 3a)  
  → CONVERSATION_CONTEXT (Phase 3 + Phase 3a – Working + Conversation Memory)  
  → BUSINESS_OBJECT_CONTEXT  
  → WORLD_CONTEXT (Phase 3a – World Intelligence Memory)  
  → TEMPORAL_CONTEXT  
  → AUTHORITY_CONTEXT (Phase 5)  
  → PRIVACY_CONTEXT (Phase 5)  
→ ReasoningContext (bounded, relevant, explainable)  
→ ReasoningLayer (original, preserved)  
→ PlannerLayer (original, preserved + Phase 13)  
→ GovernanceLayer (original, preserved + Phase 5)  
→ ExecutorLayer (original, preserved)  
→ ObserverLayer (original, preserved + Phase 14)
```

PHASE 10 — Internal-First Knowledge Resolution

Same as previous plan. KnowledgeSufficiencyEvaluator consults Phase 3a Organizational Memory freshness before external retrieval.

PHASE 11 — World Intelligence Domain

Same as previous plan. World Intelligence Memory (Phase 3a) governs retention.

PHASE 12 — Relevance Intelligence

Same as previous plan.

PHASE 13 — Persistent Plans & Governed Action

Same as previous plan. Plans persist through process restart.

PHASE 14 — Close the Learning Loop

Same as previous plan. Learning output feeds Organizational Memory (Phase 3a) and Human Preference Memory (Phase 3a).

PHASE 15 — Relationship-Aware SHUNYA Assistant

Same as previous plan. Assistant reads Phase 3a Relationship Memory + Human Preference Memory.

PHASE 16 — Persistent SHUNYA Application Shell

Same as previous plan. Shell mounts after all backend domain phases stabilize.

PHASE 17 — Mobile Application Runtime

Same as previous plan.

CORRECTED RISK REGISTER

Risk	Mitigation
Identity resolution merges wrong persons	Never silently merge. AMBIGUOUS state blocks inference.
Conversation context storage overhead	Working memory TTL; Conversation memory TTL per sensitivity
PERSONAL data exposed to wrong context	Sensitivity classification + Memory Eligibility + Visibility gates
LLM API cost	Provider-agnostic; local model fallback; deterministic routing preserved
Learning promotes false patterns	PatternCandidate validation chain; evidence threshold before promotion
World data quality unreliable	Source authority ranking; conflict detection; freshness evaluation
HTMX shell breaks pages	Workspace-fragment responses; test every route

CORRECTED ROLLBACK STRATEGY

Correction #6 applied. Three distinct concerns:

CODE ROLLBACK

- Each phase creates git tag: phase-0-baseline, phase-1-complete, etc.
- Rolling back code: git checkout phase-N-complete
- Unicorn/systemd/nginx config never modified by evolution
- Old code deploys via systemctl restart panchi

- **Compatibility window:** Phase N code continues to read Phase N+1 schemas via nullable columns + adapters

SCHEMA RECOVERY

- ALL migrations are ADDITIVE:
- New tables: CREATE TABLE (safe to have extra empty tables)
- New columns: ADD COLUMN ... NULL (existing code ignores unknown columns)
- No destructive DROP TABLE, ALTER COLUMN DROP NOT NULL, or column removal in evolution path
- Migration version tracking via Alembic (already configured)
- Pre-migration: pg_dump -h localhost -U panchi panchi_db > pre_phase_N.sql
- Rollback schema: alembic downgrade (when available); otherwise the column remains NULL — new code handles NULL gracefully

DATA RECOVERY

- Data created under new schemas survives code rollback (columns are additive, data stays)
- Dual-read period: old code reads via adapters/compatibility views; new code reads native columns
- Forward-fix strategy: if a phase introduces incorrect data, write a corrective migration (not destructive)
- No production rollback path depends on DROP TABLE
- Adapter pattern: Lead.person property returns Person object when person_id is set, falls back to customer_name string when NULL. Old templates continue working regardless.

MIGRATION VERSION TRACKING

- Alembic migration files: migrations/versions/
- Each phase adds exactly one migration
- alembic history shows full version lineage
- alembic downgrade -1 for rollback when additive-only guarantees permit
- For additive-only migrations (all of ours), downgrade is optional — old code ignores new columns

FINAL CORRECTED PHASE MAP

Phase 0 – Safety, Baseline & Architectural Convergence
Phase 1 – Unified Human Identity Domain (Person + PersonIdentity + Role Profiles)
Phase 2 – Relationship Intelligence Domain (EXPLICIT vs INFERRED)
Phase 3 – Conversation & Interaction Domain (unified message model)
Phase 3a – MEMORY ARCHITECTURE (7 memory classes; universal memory metadata)
Phase 4 – Human Context Domain (HumanSignal, PatternCandidate)
Phase 5 – Privacy, Sensitivity & Memory Eligibility (5-level classification)
Phase 6 – Evidence, Provenance & Source Intelligence
Phase 7 – Source/Evidence/Fact/Inference Runtime Distinction
Phase 8 – LLM Intelligence Runtime (provider-agnostic)
Phase 9 – Context Fusion Engine (13 context inputs → ReasoningContext)
Phase 10 – Internal-First Knowledge Resolution (sufficiency + freshness)
Phase 11 – World Intelligence Domain (external source adapters)
Phase 12 – Relevance Intelligence (organization/person/urgency evaluation)
Phase 13 – Persistent Plans & Governed Action
Phase 14 – Close the Learning Loop (candidate states → organizational memory)
Phase 15 – Relationship-Aware SHUNYA Assistant
Phase 16 – Persistent SHUNYA Application Shell
Phase 17 – Mobile Application Runtime

6 corrections applied: 1. ✓ Memory Architecture (Phase 3a) with 7 memory classes 2. ✓ Conversation Domain with Interaction model + channel mapping 3. ✓ PERSONAL sensitivity classification restored (5-level) 4. ✓ Person model with canonical_name + preferred_name + role profiles 5. ✓ Relationship model with confidence, source_type EXPLICIT vs INFERRED 6. ✓ Rollback strategy: CODE vs SCHEMA vs DATA; additive-only migrations; dual-read adapters